Year 11 Mathematics

Number 9

6.1 Percentages of Quantities

Percentages of Quantities

STARTER · 5 MIN

Date: _____ **Title:** Percentages of Quantities

LEARNING OBJECTIVES

- Find a percentage of an amount without a calculator
- Use a decimal multiplier to find a percentage
- Write one quantity as a percentage of another

STARTER *Answer in your book. No calculators.*

1. Find 10% of 80
2. Write 0.35 as a percentage
3. Write 7% as a decimal
4. Work out 0.2 × 150

Starter — Answers

STARTER

Tick or correct in green pen

1. $80 \div 10 = \mathbf{8}$

2. **35%**

3. **0.07**

4. **30**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find a percentage of an amount without a calculator	☐
SILVER	I can use a decimal multiplier to find a percentage	☐
GOLD	I can write one quantity as a percentage of another	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Percentages

INTRODUCTION · 3 MIN

Per cent means 'out of 100'. Build percentages from 10%, 5% and 1%, or use a multiplier.

10%

Divide by 10.

10% of 240 = 24

1%

Divide by 100.

1% of 240 = 2.4

Multiplier

Percentage ÷ 100.

35% → 0.35

As a percentage

Fraction × 100.

12 out of 40 = 30%

Examples 1 and 2 — Without a Calculator

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Find 35% of 240 without a calculator.

Example 2

Find 17.5% of 80 without a calculator.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

10% = 24

30% = 72

5% = 12

35% = 72 + 12

= 84

Tip: build up from 10% and 5%.

Example 2

10% = 8

5% = 4

2.5% = 2

17.5% **= 14**

Tip: halve to get 5% and 2.5%.

Task 1

PRACTICE · 9 MIN

BRONZE *No calculators*

1. 20% of 60
2. 15% of 300
3. 45% of 80
4. 1% of 650
5. 12.5% of 160

★ 0.5% of 900

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $2 \times 6 = \mathbf{12}$
2. $30 + 15 = \mathbf{45}$
3. $32 + 4 = \mathbf{36}$
4. $650 \div 100 = \mathbf{6.5}$
5. $16 + 4 = \mathbf{20}$

★ $9 \div 2 = \mathbf{4.5}$

Examples 3 and 4 — Using a Multiplier

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint multiplier = $\frac{\text{percentage}}{100}$

Example 3

Find 23% of £56.

Example 4

Find 8.5% of 420 kg.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

23% = 0.23

0.23 × 56

£12.88

Tip: divide the percentage by 100.

Example 4

8.5% = 0.085

0.085 × 420

35.7 kg

Tip: 8.5% is 0.085, not 0.85.

Task 2

PRACTICE · 9 MIN

SILVER *Use a multiplier*

Hint multiplier = $\frac{\text{percentage}}{100}$

1. 37% of £85
2. 62% of 450 m
3. 4.5% of £2400
4. 120% of 65
5. 0.8% of 3500

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $0.37 \times 85 = \mathbf{£31.45}$
2. $0.62 \times 450 = \mathbf{279\ m}$
3. $0.045 \times 2400 = \mathbf{£108}$
4. $1.2 \times 65 = \mathbf{78}$
5. $0.008 \times 3500 = \mathbf{28}$

Examples 5 and 6 — Percentage of Another and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $\% = \frac{\text{part}}{\text{whole}} \times 100$

Example 5

(2 marks)

A test has 60 marks. Sam scores 45. What percentage is this?

Example 6

(3 marks)

Ali earns £2400 a month. He pays 22% in tax, then 15% of the rest on rent. Find his rent.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

45 ÷ 60 × 100

75%

Tip: part ÷ whole × 100.

Example 6

Tax: 0.22 × 2400 = £528

Rest: 2400 − 528 = £1872

Rent: 0.15 × 1872 = £280.80

Tip: work out the rest first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $\% = \frac{\text{part}}{\text{whole}} \times 100$

1. Write 18 out of 72 as a percentage. (2 marks)

2. A class has 32 students. 14 are boys. What percentage are girls? (2 marks)

3. Write 450 g as a percentage of 2 kg. (2 marks)

4. A car costs £12 000. A 15% deposit is paid. How much is left to pay? (2 marks)

5. A shop sells 35% of 240 shirts and 40% of 150 jumpers. How many items are sold? (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $18 \div 72 \times 100 = \mathbf{25\%}$

2. $18 \div 32 \times 100 = \mathbf{56.25\%}$

3. $450 \div 2000 \times 100 = \mathbf{22.5\%}$

4. $0.15 \times 12\,000 = 1800, \quad 12\,000 - 1800 = \mathbf{£10\,200}$

5. $84 + 60 = \mathbf{144}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find 10% of 340

SILVER Find 15% of £72 using a multiplier

GOLD Write 27 out of 45 as a percentage

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $340 \div 10 = \mathbf{34}$

SILVER $0.15 \times 72 = \mathbf{£10.80}$

GOLD $27 \div 45 \times 100 = \mathbf{60\%}$

Exam Practice — Percentages Without a Calculator

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Work out 35% of 260

Do not use a calculator.

.....

Question 2 (2 marks)

Work out 12.5% of £96

Do not use a calculator.

..... £

Exam Practice — Using a Multiplier

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint multiplier = $\frac{\text{percentage}}{100}$

Question 3

(2 marks)

Work out 17% of £245

..... £

Question 4

(2 marks)

A tank holds 850 litres.

It is 6.4% full.

How many litres are in the tank?

..... litres

Exam Practice — Percentage of Another and Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $\% = \frac{\text{part}}{\text{whole}} \times 100$

Question 5

(2 marks)

Write 750 g as a percentage of 3 kg.

..... %

Question 6

(3 marks)

Maya earns £2800 a month.
She pays 20% in tax.
She spends 30% of the rest on rent.
How much does she spend on rent?

..... £

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

10% = 26, 30% = 78, 5% = 13 — M1

78 + 13 = 91 — A1

Tip: build up from 10% and 5%.

Question 2

10% = 9.60, 2.5% = 2.40 — M1

9.60 + 2.40 = £12 — A1

Tip: 12.5% is the same as one eighth.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

0.17 × 245 — M1

= £41.65 — A1

Tip: 17% = 0.17.

Question 4

0.064 × 850 — M1

= 54.4 litres — A1

Tip: 6.4% is 0.064, not 0.64.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

3 kg = 3000 g — M1

$$\frac{750}{3000} \times 100 = 25\%$$ — A1

Tip: use the same units first.

Question 6

tax = 0.2 × 2800 = £560 — M1

rest = 2800 − 560 = £2240 — M1

rent = 0.3 × 2240 = £672 — A1

Tip: 'of the rest' means after tax.